

Literature-Known Superreflexive Δ -Point Subcase With an Explicit Direct Proof

Status packet for arXiv:2301.04433

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Source Question

The source paper is Miguel Martin, Yoel Perreau, and Abraham Rueda Zoca, *Diametral notions for elements of the unit ball of a Banach space*, arXiv:2301.04433 [1].

Question 7.7 asks for isomorphic restrictions on Banach spaces containing Δ -points and the other diametral notions. In particular, it asks whether a reflexive or even super-reflexive Banach space can contain Δ -, super Δ -, ccs Δ -, Daugavet, or super Daugavet points.

Packet Status

This is a literature-known partial subcase, not a new partial result claimed by the run. The later paper arXiv:2303.00511 explicitly answers this part of Question 7.7. The purpose of this packet is to keep an additional direct proof found during the run, using a simpler explicit equivalent norm on ℓ_2 .

By now, the only *isomorphic* restriction which is known for a Banach space to contain Δ -points or even Daugavet points is that it cannot be finite-dimensional. It would be interesting to find some more.

Question 7.7. Find isomorphic restrictions for a Banach space to contain Δ -points or any of the other diametral notions. In particular, is it possible for a reflexive or even super-reflexive Banach space to contain Δ -, super Δ -, ccs Δ -, Daugavet or super Daugavet points?

The results about absolute sums in Subsection 3.2 are not complete in the case of super Daugavet points and they are even less clear in the case of ccs notions. Here are two possible questions.

Definitions Used

For $x \in S_X$, x is a super Δ -point if

$$\sup_{y \in W} \|x - y\| = 2$$

for every non-empty relatively weakly open subset W of B_X containing x . It is a ccs Δ -point if the same equality holds for every convex combination of slices of B_X containing x . Since slices are relatively weakly open, every super Δ -point is a Δ -point.

Proof Intuition

The obstruction in Question 7.7 is isomorphic: can a space still be superreflexive while its norm has diametral points? We keep the underlying linear space equal to ℓ_2 , so superreflexivity is automatic once the new norm is equivalent to the Hilbert norm.

The norm is designed so that $e_1 + e_n$ remains on the unit sphere, while the single coordinate e_n has norm 2. Since $e_n \rightarrow 0$ weakly, every weak neighborhood of e_1 in the unit ball contains some $e_1 + e_n$, and that point is exactly distance 2 from e_1 . A separate coordinate inequality makes e_1 extreme in the unit ball; once a convex combination of slices contains an extreme point, all slices in the representation must contain that point, so the same weak-null sequence proves the ccs version.

Direct Proof

Theorem 1. *There exists a superreflexive Banach space containing a point that is simultaneously a Δ -point, a super Δ -point, and a ccs Δ -point.*

Proof. Let (e_n) be the usual basis of ℓ_2 . For $x = (x_n)_{n=1}^\infty \in \ell_2$, define

$$\|x\|_N = \max \left\{ |x_1|, \left(\sum_{n=2}^\infty |x_n|^2 \right)^{1/2}, \sup_{n \geq 2} |x_1 - 2x_n| \right\}.$$

This is a norm: it is the maximum of seminorms, and the first two terms separate points. It is equivalent to the Hilbert norm. Indeed,

$$\|x\|_2 \leq \sqrt{2} \max \left\{ |x_1|, \left(\sum_{n=2}^\infty |x_n|^2 \right)^{1/2} \right\} \leq \sqrt{2} \|x\|_N,$$

while

$$\|x\|_N \leq \max \{ \|x\|_2, \|x\|_2, |x_1| + 2 \sup_{n \geq 2} |x_n| \} \leq 3 \|x\|_2.$$

Let $Y = (\ell_2, \|\cdot\|_N)$. Since Y is linearly isomorphic to Hilbert space, it is superreflexive.

We next record three elementary norm values. For every $n \geq 2$,

$$\|e_1\|_N = 1, \quad \|e_1 + e_n\|_N = 1, \quad \|e_n\|_N = 2.$$

The sequence $e_1 + e_n$ converges weakly to e_1 in Y , because the new norm is equivalent to the Hilbert norm and $e_n \rightarrow 0$ weakly in ℓ_2 . Let W be any relatively weakly open subset of B_Y containing e_1 . Then $e_1 + e_n \in W$ for all sufficiently large n , and for such n

$$\|e_1 - (e_1 + e_n)\|_N = \|-e_n\|_N = 2.$$

Thus $\sup_{y \in W} \|e_1 - y\|_N = 2$, and e_1 is a super Δ -point. In particular, e_1 is a Δ -point.

It remains to prove the ccs assertion. First we show that e_1 is an extreme point of B_Y . Suppose $u, v \in B_Y$ and

$$e_1 = \frac{u + v}{2}.$$

Since $|u_1| \leq 1$, $|v_1| \leq 1$, and $(u_1 + v_1)/2 = 1$, the equality case in the unit disc gives $u_1 = v_1 = 1$. Fix $n \geq 2$. From $u, v \in B_Y$ we have

$$|1 - 2u_n| \leq 1, \quad |1 - 2v_n| \leq 1,$$

and from $(u_n + v_n)/2 = 0$ we have $v_n = -u_n$. Hence

$$|1 - 2u_n| \leq 1 \quad \text{and} \quad |1 + 2u_n| \leq 1.$$

The two closed discs $\{z : |1 - 2z| \leq 1\}$ and $\{z : |1 + 2z| \leq 1\}$ are tangent at 0, so $u_n = 0 = v_n$. This holds for every $n \geq 2$, and therefore $u = v = e_1$. Thus $e_1 \in \text{ext } B_Y$.

Now let

$$C = \sum_{i=1}^m \lambda_i S_i$$

be a convex combination of slices of B_Y , with $\lambda_i > 0$ and $\sum_i \lambda_i = 1$, and suppose $e_1 \in C$. Choose $z_i \in S_i$ such that $e_1 = \sum_i \lambda_i z_i$. Since e_1 is an extreme point of the convex set B_Y , every z_i must equal e_1 . Thus $e_1 \in S_i$ for each i . The finite intersection

$$W = \bigcap_{i=1}^m S_i$$

is a relatively weakly open subset of B_Y containing e_1 . Therefore some $e_1 + e_n$ belongs to W . For this n ,

$$e_1 + e_n = \sum_{i=1}^m \lambda_i (e_1 + e_n) \in C,$$

and, as above, $\|e_1 - (e_1 + e_n)\|_N = 2$. Hence

$$\sup_{y \in C} \|e_1 - y\|_N = 2.$$

So e_1 is a ccs Δ -point. □

Limitations

This is a partial answer to Question 7.7. It proves that superreflexive spaces can contain Δ -, super Δ -, and ccs Δ -points. It does not prove that a reflexive or superreflexive space can contain a Daugavet point or a super Daugavet point.

Novelty and Provenance Check

The result itself is already known in later literature: Abrahamsen–Aliaga–Lima–Martiny–Perreau–Prochazka–Veeorg [2] explicitly cite the MPRZ question and construct a superreflexive renorming of ℓ_2 with super Δ - and ccw Δ -points. The present packet is therefore not counted as a new partial result. Its purpose is to record a direct proof found during the run, using the explicit norm above rather than invoking the later theorem.

The run indexes were checked for arXiv:2301.04433, superreflexive Δ -points, ccs Δ -points, and the existing arXiv:2303.00511 literature-status packet. This packet should be read together with that provenance packet.

Human Review Recommendation

Review as a literature-known partial subcase with an additional direct proof. The most important checks are the equivalence of $\|\cdot\|_N$ with the Hilbert norm, the weak convergence $e_1 + e_n \rightarrow e_1$, the extremality proof for e_1 , and the final reduction from a ccs containing e_1 to a finite intersection of slices containing e_1 .

References

- [1] M. Martin, Y. Perreau, and A. Rueda Zoca, *Diametral notions for elements of the unit ball of a Banach space*, arXiv:2301.04433, 2023.
- [2] T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511, 2023.